Project Development Guidelines

Objective

This document contains **3 full-stack project requirements (PRDs)** designed to help learners apply their technical knowledge through practical development.

Instructions for Learners

1. **Choose One Project**
    - Review all three project descriptions provided in this document.
    - Select **any ONE project** based on your interest.
    - Each learner is required to develop only one project.
2. **Analyze the Complete PRD**
    - Read and understand the selected project's Product Requirements Document (PRD).
    - Analyze the objectives, user roles, functional modules, user stories, business rules, and acceptance criteria.
    - Identify the required frontend screens, backend APIs, and database tables.
3. **Plan Before Development**
    - Prepare the database design and application architecture.
    - Identify the modules and their dependencies.
    - Plan the development tasks and implementation sequence.
4. **Develop the Application**
    - Implement the selected project using your trained technology stack:
        - **Java Track:** Java, Spring Boot, MySQL, Angular.
        - **.NET Track:** C#, ASP.NET Core Web API, SQL Server/MySQL, Angular.
    - Follow proper coding standards and a layered architecture**(Micro services Must)**
    - Implement all mandatory functional modules mentioned in the PRD.
5. **Test and Validate**
    - Test all major application features.
    - Validate user roles, authentication, business rules, and API functionality.
    - Ensure the application meets the PRD acceptance criteria.
6. **Final Submission**
    - Complete source code in a GitHub repository.
    - Database scripts and sample data.
    - README with project setup instructions.
    - API documentation.
    - Screenshots of completed modules.
    - Working project demonstration.

Important Guidelines

- **Only ONE project must be selected and completed.**
- The complete PRD must be analyzed before starting development.
- All mandatory modules and core business requirements must be implemented.
- Learners must write, understand, and be able to explain their code.
- The final application must have a working frontend, backend, and database integration.
- Project completion will be evaluated based on functionality, code quality, database design, UI/UX, testing, and demonstration.

Expected Outcome: A fully functional, end-to-end full-stack application developed independently according to the selected PRD.

Project 01 — Employee Management & Leave Tracking System

1. Project Overview

Develop a web-based Employee Management & Leave Tracking System that enables organizations to manage employee information, attendance, and leave requests.

The system should support role-based access, automated leave balance calculations, and basic employee dashboards.

2. Project Objectives

- Digitize employee leave requests and approvals.
- Track attendance and leave balances.
- Provide dashboards for employees, managers, and HR.
- Ensure secure, role-based access to organizational data.

3. User Roles & Permissions

Feature	Admin / HR	Manager	Employee
Manage Employees	Yes	View Team	Own Profile
Mark Attendance	Yes	Own	Own
Apply for Leave	Yes	Yes	Yes
View Leave Balances	All	Team	Own

4. Functional Modules

Module 1: Authentication & Authorization

- Login using email and password.
- JWT-based authentication.
- Role-based access control.
- Password reset and logout.
- Secure route protection in Angular.

Module 2: Employee Management

- Add, view, update, and deactivate employees.
- Maintain employee ID, name, email, phone, designation, joining date, and department.
- Assign employees to managers.
- Search, filter, and paginate employee records.
- Allow employees to update permitted profile details.

Module 3: Attendance Management

- Employee check-in and check-out.
- Automatically record date and working hours.
- View daily and monthly attendance.
- HR can review and correct attendance with an audit reason.
- Prevent duplicate check-ins for the same day.

Module 4: Leave Management

- Configure leave types (Casual, Sick, Earned).
- Assign annual leave balances.
- Apply for leave with dates, type, and reason.
- View pending, approved, and rejected requests.
- Automatically deduct leave balance upon approval.
- Prevent overlapping requests and insufficient leave balances.

Module 5: Dashboard & Reports

Employee Dashboard

- Attendance summary.
- Available leave balance.
- Recent leave requests.

Admin / HR Dashboard

- Total active employees.
- Department-wise employee count.
- Monthly attendance statistics.
- Leave usage reports.

5. User Stories

ID	User Story	Priority
US-01	As a user, I want to log in securely based on my role.	High
US-02	As HR, I want to create and manage employee records.	High
US-03	As an employee, I want to check in and check out daily.	High
US-04	As an employee, I want to apply for leave and track its status.	High
US-05	As an employee, I want to view my remaining leave balance.	High
US-6	As HR, I want to configure leave types and allocations.	Medium

6. Business Rules

1. Every employee must have a unique employee ID and email.
2. Only Admin/HR can create, deactivate, or assign employees.
3. Leave requests must have valid dates and sufficient available balance.
4. Approved leave cannot be modified directly; cancellation requires a separate action.
5. Rejected leave must not reduce leave balance.
6. Employees must not access other employees' private records.
7. Attendance records must be unique per employee per date.

7. Non-Functional Requirements

Category	Requirement

Security	JWT authentication, BCrypt password hashing, role-based API authorization
Performance	Standard API responses targeted under 2 seconds for normal test loads
Usability	Responsive Angular interface with clear validation messages
Data Integrity	MySQL foreign keys, constraints, and transactional updates
Maintainability	Layered Spring Boot architecture: Controller, Service, Repository
Error Handling	Standardized HTTP status codes and meaningful error messages
Documentation	Swagger/OpenAPI documentation and project README
Testing	Unit tests for business logic and integration tests for critical APIs
Auditability	Record who approved or rejected leave and when

8. Expected Deliverables

Learners must submit:

- Angular frontend source code.
- Spring Boot backend source code.
- MySQL database schema and sample data.
- REST API documentation.
- GitHub repository with a setup README.
- Test cases and execution results.
- Screenshots of major modules.
- A working end-to-end application demonstration.

9. Project Acceptance Criteria

The project is considered complete when:

- All three roles can log in and access only authorized features.
- HR can manage employees and departments.
- Employees can record attendance and request leave.
- Managers can approve or reject team leave requests.
- Leave balances update correctly without duplicate deductions.
- Dashboards display accurate data from MySQL.
- All major workflows function through the Angular frontend and Spring Boot REST APIs.
- The application runs successfully using the provided setup instructions.

Project 02 — Internal Company Social Media Platform

1. Project Overview

Develop an Internal Company Social Media Platform that enables employees to communicate, share updates, celebrate achievements, and collaborate within their organization.

The application should function as a private corporate social network, supporting posts, comments, reactions, and company announcements.

2. Project Objectives

- Improve internal employee communication.
- Provide a centralized platform for company announcements.
- Encourage knowledge sharing and employee engagement.
- Enable employees to interact through posts, comments, and reactions.
- Maintain secure, organization-only access.

3. User Roles & Permissions

Feature	Admin / HR	Group Moderator	Employee
Manage Users	Yes	No	No
Create Posts	Yes	Yes	Yes
Edit / Delete Posts	All	Group Posts	Own Posts
Comment & React	Yes	Yes	Yes
Publish Announcements	Yes	No	No

4. Functional Modules

Module 1: Authentication & User Profiles

- Secure login using email and password.
- JWT authentication and role-based authorization.
- Employee profiles with name, designation, department, and profile picture.
- Edit profile information.
- Search employees by name or department.

Module 2: Social Feed & Posts

- Create text posts with optional images.
- View company-wide posts in a feed.
- Edit and delete own posts.
- Display author, timestamp, and engagement counts.
- Paginate posts with newest first.

Module 3: Comments & Reactions

- Add comments to posts.
- Edit or delete own comments.
- React to posts using Like.
- Remove an existing reaction.
- Display total reactions and comments.
- Prevent duplicate likes by the same user.

Module 4: Company Announcements

- Admin/HR can publish official announcements.
- Mark announcements as important.
- Display important announcements at the top of the feed.
- Allow employees to view announcement history.
- Track which employees have acknowledged announcements.

Module 5: Notifications

- Notify users when someone comments on their post.
- Notify users when their post receives a reaction.
- Notify employees about new company announcements.
- Display unread notification count.
- Mark notifications as read.

In-app notifications using database records and REST APIs are sufficient. Real-time WebSocket notifications are optional.

Module 6: Dashboard & Reports

Employee Dashboard

- Latest company posts.
- Important announcements.
- Recent notifications.

5. User Stories

ID	User Story	Priority
US-01	As an employee, I want to log in securely.	High
US-02	As an employee, I want to create and share posts.	High
US-03	As an employee, I want to comment and react to posts.	High
US-04	As an employee, I want to update my profile.	High
US-05	As HR, I want to publish company announcements.	High
US-06	As an employee, I want to receive activity notifications.	Medium
US-07	As an employee, I want to search posts and colleagues.	Medium

6. Business Rules

1. Only active employees can access the platform.
2. Every user must have a unique email address.
3. Users can edit or delete only their own posts and comments, except authorized moderators and admins.

4. Only Admin/HR can publish official company announcements.
5. Each user can have only one Like reaction per post.
6. Deleted posts must no longer appear in feeds.
7. Notifications must be associated with valid users and activities.
8. Uploaded images must be validated for file type and size.

7. Non-Functional Requirements

Category	Requirement
Security	JWT, BCrypt, role-based authorization
Performance	Standard API responses targeted under 2 seconds
Usability	Responsive social feed and intuitive navigation
Data Integrity	MySQL relationships, unique constraints, foreign keys
Maintainability	Controller, Service, Repository architecture
File Handling	Secure image upload with configurable size limits
Error Handling	Consistent API errors and frontend validation
Documentation	Swagger/OpenAPI and README
Testing	Unit and integration tests for posts, permissions, and reactions

Privacy	Company-only access and group-level visibility

8. Expected Deliverables

- Angular frontend application.
- Spring Boot backend REST APIs.
- MySQL database schema with sample data.
- API documentation.
- GitHub repository and setup README.
- Test cases and results.
- Screenshots of major modules.
- Working end-to-end demonstration.

9. Project Acceptance Criteria

The project is complete when:

- Employees can log in and manage their profiles.
- Users can create, edit, delete, and view posts.
- Comments and reactions work correctly.
- HR can publish company announcements.
- Group access permissions are enforced.
- Moderators can manage assigned group content.
- Notifications display relevant activities.
- Admin dashboards display accurate engagement metrics.
- The Angular frontend communicates successfully with Spring Boot APIs and MySQL.

03 — Team Project & Task Management System

1. Project Overview

Develop a Team Project & Task Management System that enables organizations to plan projects, assign tasks, track progress, and collaborate effectively.

The application should provide basic project management, team member assignment, task creation and assignment, and task completion tracking.

2. Project Objectives

- Centralize project and task management.
- Improve team collaboration and accountability.
- Track task progress and deadlines.

3. User Roles & Permissions

Feature	Admin	Project Manager	Team Member
Manage Users	Yes	No	No
Create Projects	Yes	Yes	No
Edit Project Details	All	Own Projects	No
Add Project Members	Yes	Own Projects	No
Create & Assign Tasks	Yes	Own Projects	Assigned Projects

Update Task Status	Yes	Own Projects	Assigned Tasks
Add Task Comments	Yes	Yes	Project Members

4. Functional Modules

Module 1: Authentication & Authorization

- Secure login using email and password.
- JWT-based authentication.
- Role-based access control.
- User profile management.
- Angular route guards and backend API authorization.

Module 2: Project Management

- Create and edit projects.
- Maintain project name, description, start date, end date, and status.
- Assign a project manager.
- Add or remove project members.
- Display a list of projects accessible to the logged-in user.
- View project summary and progress.

Module 3: Task Management

- Create, edit, and delete tasks.
- Assign tasks to project members.
- Define task title, description, priority, status, and due date.
- Support task statuses: To Do, In Progress, In Review, Done.

Module 6: Team Collaboration

- Add comments to tasks.
- Allow project members to view relevant task discussions.
- Display task update timestamps.
- Task completion summary.

5. User Stories

ID	User Story	Priority
US-01	As a user, I want to log in securely.	High
US-02	As a manager, I want to create projects and assign members.	High
US-03	As a team member, I want to view tasks assigned to me.	High
US-04	As a manager, I want to create and assign tasks.	High
US-05	As a team member, I want to update task status.	High

6. Business Rules

1. Every project must have a unique project identifier.
2. Every task must belong to a valid project.
3. Tasks can be assigned only to members of the respective project.
4. Only authorized users can create, update, or archive projects.
5. Team members can update the status of tasks assigned to them.
6. Completed tasks must remain available in project history.
7. Team members can mark only their assigned tasks as Completed.

7. Non-Functional Requirements

Category	Requirement
Security	JWT authentication, BCrypt, project-level authorization
Performance	Standard API responses targeted under 2 seconds

Usability	Responsive UI with drag-and-drop Kanban board
Data Integrity	MySQL foreign keys, transactions, and constraints
Maintainability	Layered Spring Boot architecture
Scalability	Server-side pagination, filtering, and indexed queries
Error Handling	Standard API errors and frontend validation
Documentation	Swagger/OpenAPI and README
Testing	Unit and integration tests for project permissions and task workflows
Auditability	Task status and assignment change history

8. Expected Deliverables

- Angular frontend source code.
- Spring Boot backend source code.
- MySQL database schema and sample data.
- REST API documentation.
- GitHub repository with README.
- Test cases and results.
- Screenshots of project, task, and Kanban modules.
- Working end-to-end demonstration.

9. Project Acceptance Criteria

The project is complete when:

- Users can log in and access authorized projects.

- Managers can create projects and manage project members.
- Tasks can be created, assigned, updated, and tracked.
- Project-level access restrictions are enforced.
- All major workflows operate end-to-end through Angular, Spring Boot, and MySQL.